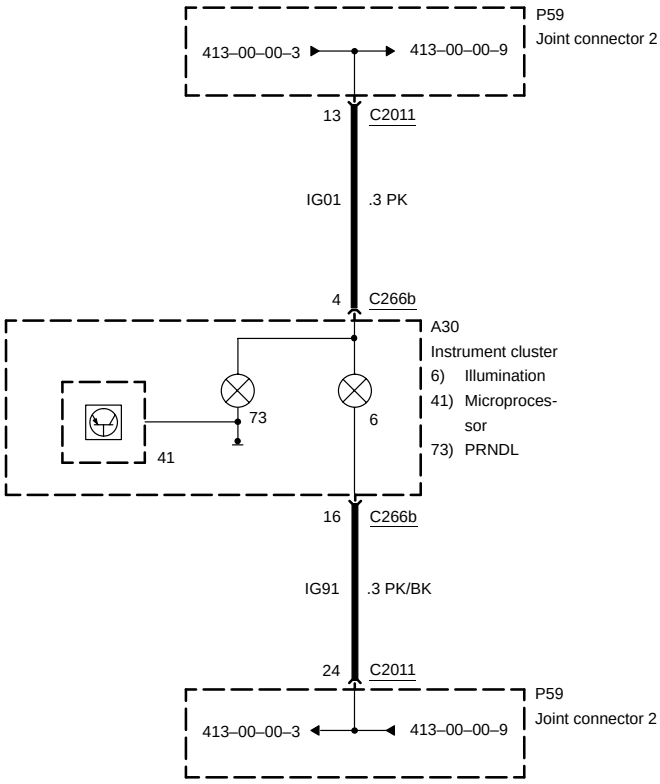conventional cluster -4